Data Management (2025/2026) - Project Proposal

Group Members

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Chosen Type of Project

[NoSQL] DBMS comparison. Select a dataset, a graph database tool (Neo4j), and a relational DBMS (PostgreSQL), and compare the results of analyses on different systems, highlighting advantages and disadvantages of each approach in terms of efficiency.

Dataset Description

Name: arXiv Dataset (filtered subset for `cs.DB` and `cs.AI`)
Reference Link: `https://www.kaggle.com/datasets/Cornell-University/arxiv`
Description: The dataset contains metadata on scholarly articles, including titles, authors, categories, and references. In order ro guarantee a highly connected network while maintaining manageable execution times, we will extract a dense subset out of the original dataset. This subset is restricted to the Database (`cs.DB`) and Artificial Intelligence (`cs.AI`) categories, resulting in approximately 185,000 papers and 313,000 unique authors.

Brief Description of the Work

The goal of this project is to model the arXiv publication and co-authorship network in both PostgreSQL and Neo4j to rigorously compare their efficiency.
After generating clean CSV files regarding the subsets, we will populate a normalized relational schema in PostgreSQL and a property graph in Neo4j. We will then benchmark the execution times and evaluate the query complexity (SQL vs Cypher) across three specific scenarios:

1. **Aggregation (SQL advantage):** Finding the most prolific authors in specific timeframes.
2. **Deep Traversal / Pattern Matching (Graph advantage):** Discovering papers cited by papers that cite a specific target (indirect citations), or finding 2nd/3rd degree co-authors.
3. **Shortest Path:** Calculating the shortest co-authorship path between two researchers.

In order to ensure a methodologically sound evaluation, the comparison will heavily focus on the impact of indexing within the relational model (Fair vs. Unfair comparison).